



Document Management System



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Title: Treatment of acute painful crisis in patients with Sickle cell disease

Purpose:
Management of uncomplicated painful Vaso-Occlusive crisis in patients with sickle cell disease.

Scope:
This document applied to all physicians pain management team involved in the management of sickle cell disease.

Management of sickle cell pain in Emergency Department:

1. Take a detailed clinical History, including potential precipitants and source of pain, previous treatment and drug history.
2. Exclude non-SCD causes of pain (e.g. surgical cause).
3. Assess for signs of other SCD complications such as stroke, acute chest syndrome, fever, splenic sequestration, and priapism.
4. Check Vital signs (temperature , BP, pulse, oxygen saturation on air, and respiratory rate).
5. Chart Pain score according to the pain assessment tool.
6. Pain assessment should be every 15-30 min for the first two hours of presentation and 4 hourly if stable. Further assessment to be done when indicated by patient's condition using approved pain score as recommended by Royal hospital policy.
7. Start IV access and with draw CBC, Reticulocyte count, UE1 and LFT.(Avoid cannulating legs and feet to avoid the increased risk of thrombosis.).
8. Analgesia should be given with 30 min of presentation. Do not delay analgesic treatment while attempting to secure venous access, if difficult IV access give Morphine 2mg s/c and avoid all I.M injection.
9. Give a bolus of Morphine sulfate 2 mg IV to all patients with severe or moderate pain base on pain score.
10. Reassess pain score and vital signs every 15-30 min, if pain score >3 repeat morphine IV dose of 2 mg. Reassess pain

11. Start Opioid IV Patient controlled analgesia (PCA) (Morphine or Fentanyl) infusion at the following setting: No Background is required to be started in ER, Background will be started in the ward if required, Morphine bolus 1mg, Fentanyl bolus 20 microgram, and lockout 5-10 minutes. Concentration fixed of (Morphine 1ml/1ml;Fentanyl 50 mcg/1ml).
12. If PCA not available then Start Morphine infusion (1:1 dilution at rate of 1mg/hr) and for break through pain give 1mg every 15 min as required.
13. Admit the patient if pain uncontrolled (with 4 hrs.)
14. Start IVF: Dextrose 5% in 1/2 NS at 125 ml/hour.
15. For synergistic effect give Paracetamol 1 gram IV and Diclofenac sodium 50 mg PO or Ketorolac 30 mg IV TID.
16. Pethidine 50-100mg IM 2-4 hourly PRN is only to be administered if the patient has proven severe allergy/side-effect to all other forms of opioid analgesia. There is proven evidence that there is an increased risk of fits and dependence.

Inpatient care:

1. Continue Opioid infusion and iv fluid
2. Insure the patient is in the following Adjuvant therapy:
 - ☐ Paracetamol 1 gram IV TID. Maximum 4 grams in 24 hours
 - ☐ Ibuprofen 400mg TID or Diclofenac sodium 50 mg TID PO or Ketorolac 30 mg IV every 8 hours (Maximum of 90 mg/day, Maximum 3 days for the treating physicians other than the pain management team). Check creatinine on day 1 and day 3 if starting Ketorolac).
 - ☐ Prescribe Anti-emetic.
 - ☐ Lactulose 15-30 mls PO HS regular.
 - ☐ Anti-histamines (for pruritus)PRN.
 - ☐ Folic acid 5mg OD.
 - ☐ Supplemental oxygen for SPO2 <95% on room air.
 - ☐ Low molecular weight heparin at prophylactic dose OD subcutaneously.
 - ☐ Incentive spirometry for patients with chest or back pain.
3. Other Adjuvant therapy:
 - ☐ Naloxone should be available on the ward for treatment of opioid excess (severe respiratory depression (RR<10/min) or sedation score is 3 or 4) and to be given by medical team.
 - Naloxone dose: Dilute 100 microgram in 1 ml IV with repeat boluses after 2-3 minutes if required till patient breath normally. Max dose of Naloxone is 1.5 - 3 mcg/kg.
 - Refer the patient to Anesthetists for further assessment.
 - For Naloxone infusion to refer the patient to HD/ICU.
4. Consult pain management team if pain score >3.
5. Monitor the patient vital signs, O2 Saturation, pain score and sedation score every 15-30min until pain is controlled and patient is stabilized then every 4 hours x2 then routine thereafter.
6. If Patient documented to have allergic to morphine then use Fentanyl,Suitable bolus dose of fentanyl is: 20 microgram IV which can be repeated after 10min if pain persist then start fentanyl infusion 15 mic/hr.
7. Pethidine is not recommended because of risk of seizures at high doses.
8. Women with SCD present with pregnancy is safe to use parenteral opioids during painful crisis. **Avoid** NSAIDs, Tramadol, Pregabalin and Gabapentine due to possible risks of fetal abnormalities.

- Conscious, appropriately responsive and able to obey commands.
- Respiratory rate >10/min (Normal RR 12-20/min) and O2 saturations on room air >94%.
- SpO2 saturations >95% on oxygen therapy.
- Verbal consent to be taken from patient prior starting PCA.
- Patient is able to use the handset effectively.
- Patients must not leave the ward whilst on PCA.

Special Precautions:

- Patient is confused, unconscious, and unable to obey commands or has a high level of anxiety.
- Opioids for parenteral use that may be contra-indicated/caution in any specific patients e.g. allergy or respiratory disease; sleep apnoea, acute asthma attack or severe COPD. Seek respiratory medical advice if opioids are required.
- Patient refuses or unable to use PCA or parenteral/ s/c injections.
- Use parenteral opioids with caution in patients with moderate to severe renal or liver impairment; discuss with the ward pharmacy and the pain team for further advice.

PCA regimes:

Regime Fentanyl Morphine

Concentration of fentanyl 50mcg/1ml , morphine 1mg/1ml.

Background Continuous Infusion rate 0-15mcg 1-2mg.

Lockout time 5-10 min 5-10min.

PCA bolus dose range 15-20 mcg 1-2mg.

When to stop PCA?

When respiratory complication occurs RR falls <10/min; or Oxygen saturation in room air are <92 %.
Increase in RR of >24/min can be related to Acute Chest Syndrome(ACS) .

Sedation score:

- ☐ To be recorded every 15-30 min for the first 2 hours then 4 hourly unless condition changed.
- ☐ If sedation score is 3-4 , to stop PCA and call for help.

Pasero Opioid – induced Sedation scale (POSS) (Modified for AACA)

S = Sleep, easy to arouse

Action: Awaken patient to determine arousability (1-4 below) before administering a PCA bolus dose.

1 = Awake and alert

Action: Acceptable; may administer PCA bolus dose.

2 = Slightly drowsy, easily aroused

Action: Acceptable; may administer PCA bolus dose.

3 = Frequently drowsy, arousable, drifts off to sleep during conversation.

Action: Unacceptable; notify nurse immediately.

4 = Somnolent, minimal or no response to verbal and physical stimulation.

Action: Unacceptable; notify nurse immediately.

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